



Low Power Wide Area (LPWA) networks evolution

Abstract:

Machine-to-machine wireless interconnectivity is not always about high bandwidth connections operating at latencies of milliseconds. LPWA networks exist to service a segment of devices that consume low data rates, require very high battery life, and are installed in locations hard to reach. This whitepaper traces the evolution of LPWA technologies from 2.5G modules, to technologies using unlicensed spectrum, to mainstream licensed access via 3GPP technologies.

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Table of Content

What is LPWA?	4
Early deployments	5
Technologies in unlicensed spectrum	6
LTE-M	7
NB-IoT	7
5G NR-Light	8
Conclusion	8
About the Author	9
About Sasken	9



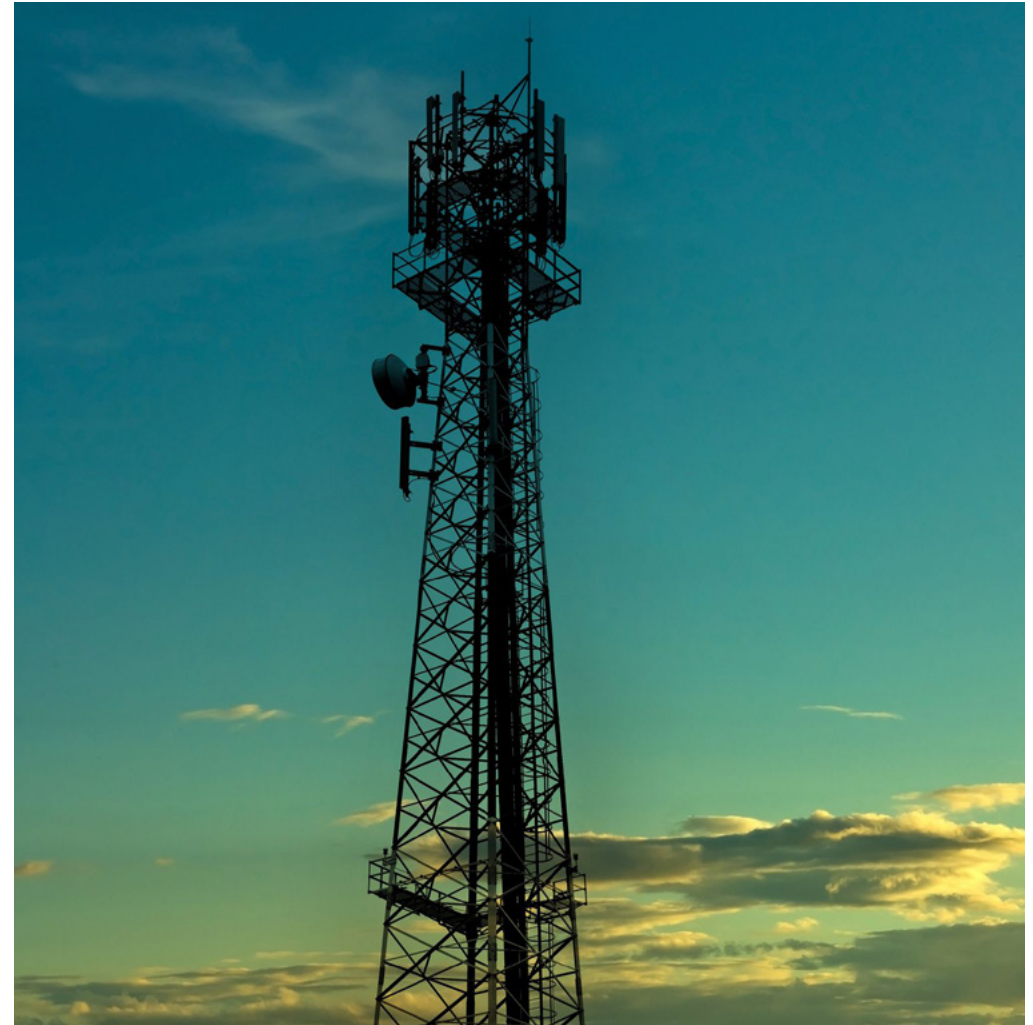
What is LPWA?

Wireless access is commonly associated with voice, messaging, social networking, and the growing demand for broadband speeds. The promise of 2G and 3G has eventually been delivered with 4G/LTE, where bandwidths comparable to wired broadband are delivered over the air. 5G takes this further to Gigabit speeds and latency in milliseconds, enabling a wide range of machine-to-machine applications. These technologies primarily address the need for providing affordable-to-premium services to thousands of users in a locality, where charges are linked to bandwidth consumed.

There is a whole family of use cases that require a different network implementation profile:

- Devices that consume very low quantities (few bytes to 100+ kilobits) of data, with long idle periods between data bursts
- Data which must be reliably delivered to and from devices located in basements (e.g., electricity meters)
- Devices located at humanly inaccessible locations (e.g., meters in industrial premises) and demanding very long battery life (e.g., 10 years with a standard AA-type battery)
- For financial viability, the outdoor transmitters (or cell sites) must support cells with an extensive range (10+ km)

Networks supporting the above categories of use cases are called Low-Power Wide Area (LPWA) networks. Low power indicates the need for devices to have low power consumption to enable long battery life. Wide Area indicates the large area that the network nodes must cover to support large numbers of low-bandwidth devices.



Early deployments

Early solutions depended on available and affordable wireless solutions for applications such as taxi meters and asset tracking. The obvious choices were 2.5 G cellular-based modules that run on (E)GPRS and CDMA. While these might not have been as cost-effective as desired, they drove adoption in large enough volumes to prove the relevance of low-bandwidth connectivity. The adoption of these 2G technologies has been so good that several countries are deferring the shutdown of their 2G networks beyond the shutdown of 3G networks.

While cellular operators took notice of the slow but steady uptake of low-bandwidth applications using 2.5G technologies, the challenge remained to deliver the promise of broadband over wireless. Social networking and the advent of smartphones and tablets have increased the demand for wireless data. But wireless broadband connectivity remains the key to realizing revenues. This kept cellular operators, and by extension, 3GPP, focused on pursuing LTE/4G. This led to innovation by an alternate set of industry players.



Technologies in unlicensed spectrum

With the high cost of spectrum worldwide for wireless applications, licensed spectrum has become the domain of cellular operators. With 4G, all cellular operators worldwide came together under the aegis of 3GPP, to deploy 4G/LTE based technologies in licensed spectrum.

This led to wireless innovation in unlicensed spectrum, where no hefty up-front investments for spectrum are required. The catch here is that available spectrum is quite limited, thereby limiting the available network capacity.

Sigfox is one LPWA technology developed by the French operator of the same name. Sigfox operates in license-free sub-GHz frequencies permitting up to 140 messages a day, each of which can be up to 12 octets, and transmitted at a speed up to 100 bits per second. The technology has been deployed in at least 72 countries.

LoRa (for LOng RANGE) is another LPWA technology which operates on license-free sub-GHz frequencies as well as 2.4 GHz, supporting data rates ranging from 0.3 Kbps to 27 Kbps.

With cellular operators focusing on 4G for licensed spectrum deployments, both LoRa and SigFox got an early start in deploying LPWA networks. But as demand grows and more devices are deployed, the limited availability of unlicensed spectrum may become a bottleneck.



LTE-M

Early LTE deployments also focused on delivering higher bandwidth in the 100s of Mbps. To be optimized for higher bandwidth, the minimum RF bandwidth that any device should receive was pegged at 20 MHz. This would prove costly for low power devices that were to be deployed in massive volumes.

As part of 3GPP Release 13, the need for LPWA support commenced. The outcome was the specification of Category-M1 LTE devices. These devices would be supported by special carriers that co-exist with regular LTE services. The RF bandwidth required for operation was brought down to 1.4 MHz, all the way down from 20 MHz for regular LTE devices. Download speeds for this category of devices have been restricted to 1 Mbps. These specifications reduce the cost of devices, encouraging large-scale deployments. This alone was not sufficient for the spectrum of devices consuming the lowest bitrate.

NB-IoT

Also in 3GPP Release 13, another category of devices were specified as Category-NB1 – which goes by the popular name of Narrowband IoT or NB-IoT. These would operate with an RF carrier bandwidth of 180 KHz. Supported data rates ranged from 50 bps to 200 Kbps. By repetitive transmission of data, the signals penetrating to basement levels would be received reliably. NB-IoT also enables a battery life of 10 years for devices.

This enabled devices which cost 25% of, and would serve more use cases than, traditional GPRS/CDMA-based devices. As with the case of LTE-M, NB-IoT transmissions would also co-exist with LTE deployments.



5G NR-Light

As was the case in LTE, 5G NR specifications in 3GPP also followed the path of pursuing higher bandwidth for Gigabit data speeds. Yet, the co-existence of LTE-M and NB-IoT deployments with 5G NR deployments were factored in right from 3GPP Release 15, where 5G was introduced.

3GPP Release 17 has introduced NR-Light or RedCap (for Reduced Capability) for mid-tier IoT applications, above what is currently provided by LTE-M and NB-IoT. NR-Light can co-exist with LTE-M and NB-IoT deployments, and devices operate on a 20 MHz RF carrier bandwidth. Data transfer speeds can go as high as 10 Mbps. Reliability and latency will be better than LTE-M and NB-IoT. NR-Light will enable low-complexity, low-cost devices with extended battery life when compared to devices with similar throughput.

NR-Light has been designed to enable use cases currently not served by LTE-M and NB-IoT. Devices will be available from 2024. Together with LTE-M and NB-IoT, NR-Light is anticipated to provide extensive capacity and coverage for low-power IoT devices using licensed spectrum. This gives an edge to these 3GPP technologies over competing technologies that operate in unlicensed spectrum.

Conclusion

With delays in 3GPP standardization and deployment, LoRa and Sigfox have taken a head start with deployments. With lower pricing and the absence of spectrum costs, service deployments with high volumes of devices have taken place worldwide. Sigfox filed for bankruptcy in 2022, despite widespread adoption of its technology.

Deployments based on 3GPP technologies are also growing. Larger capacities, better coverage, and secure data transfer are differentiators that these bring to the customer. The ongoing sunset of 2G/3G networks is also driving migration to NB-IoT and LTE-M. With NB-IoT, LTE-M, and NR-Light, 3GPP technologies cover a wide spectrum of low-power applications.

With more than 25 years of experience in ETSI/3GPP technologies, Sasken is assisting its customer base to adopt 3GPP technologies for the Internet of Things. With deep expertise in cellular modem engineering and integration, Sasken helps customers integrate cellular connectivity into their products. With deep expertise across cellular modems, radio access networks, and cloud-based applications, Sasken is well equipped to assist customers with adopting LPWA technologies for their tailored use cases.



About the Author

Roney Paul Kurien is an industry veteran who started his career at Sasken (then Silicon Automation Systems) in 1996. Beginning with CDMA Radio Control SW, he went on to play a pivotal role in the development of Sasken's widely deployed WCDMA UE protocol stack. Roney also had successful stints with several multinationals where his responsibilities ranged from device deployments into Tier-1 operator networks to development and commercialization of 3G and 4G small cells. Roney also played a leadership role in establishing the wireless practice at a highly successful startup, delivering turn-key solutions. Roney is responsible for the roadmap, solutions, and partnerships for Sasken's Wireless business portfolio.

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 32 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.





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